

IPC144 – Introduction to C Programming Practice Exercises

Week 13 – Working With Files

Q1. (Write Statements to Accomplish a Task) Write statements that accomplish each of the following. Assume that the structure

```
struct person {  
    char lastName[15];  
    char firstName[15];  
    char age[4];  
};
```

has been defined and that the file is already open for writing.

- Initialize the file "nameage.dat" so that there are 100 records with lastName = "unassigned", firstName = "" and age = "0".
- Input 10 last names, first names and ages, and write them to the file.
- Update a record; if there's no information in the record, tell the user "No info".
- Delete a record that has information by reinitializing that particular record.

Q2. (Hardware Inventory) You're the owner of a hardware store and need to keep an inventory that can tell you what tools you have, how many you have and the cost of each one. Write a program that initializes the file "hardware.dat" to 100 empty records, lets you input the data concerning each tool, enables you to list all your tools, lets you delete a record for a tool that you no longer have and lets you update any information in the file. The tool identification number should be the record number. Use the following information to start your file:

Record #	Tool name	Quantity	Cost
3	Electric sander	7	57.98
17	Hammer	76	11.99
24	Jig saw	21	11.00
39	Lawn mower	3	79.50
56	Power saw	18	99.99
68	Screwdriver	106	6.99
77	Sledge hammer	11	21.50
83	Wrench	34	7.50

Q3. Develop a small airline reservation system. The database of flight information should be kept in a file of structures with the following components:

- a) Flight number (including airline code)
- b) City of departure
- c) Destination
- d) Date and time of departure
- e) Date and time of arrival
- f) Number of first-class seats still available
- g) Number of first-class seats sold
- h) Number of coach seats still available
- i) Number of coach seats sold

Include in your program separate functions for creation, deletion, and update of flight records. Also, implement `make_reservation` and `cancel_reservation` functions.

Q4. Write a program that helps the user to consider a range of interest rates for a mortgage over 20, 25, and 30 years. Prompt the user to enter the amount of the loan and a minimum and maximum interest rate (in whole percentages).

Then write a text file containing a table of the form,

Loan Amount: \$50,000.00

Interest Rate	Duration (years)	Monthly Payment	Total Payment
10.00	20	_____	_____
10.00	25	_____	_____
10.00	30	_____	_____
10.25	20	_____	_____
		.	
		.	
		.	

The output file produced should contain payment information on a particular loan amount for interest rates from the minimum rate to the maximum rate in increments of 0.25 percent. The loan durations should be 20, 25, and 30 years.

Output the monthly payment and total payment values rounded to two decimal places. You may neglect the fact that because the monthly payment must be rounded, the final payment will be slightly different.

Q5. Cooking recipes can be stored on a computer and, with the use of files, can be quickly referenced.

a. Write a function that will create a text file of recipes from information entered at the terminal. The format of the data to be stored is

1. recipe type (dessert, meat, etc.)
2. subtype (for dessert, use cake, pie, or cookies)
3. name (e.g., German chocolate)
4. number of lines in the recipe to follow
5. the actual recipe

Item 3 should be on a separate line.

b. Write a function that will accept as parameters a file and a structured record of search parameter bounds. The function should display all recipes satisfying the search parameters.

Q6. (Telephone-Number Word Generator) Standard telephone keypads contain the digits 0–9. The numbers 2–9 each have three letters associated with them, as is indicated by the following table:

Digit	Letter	Digit	Letter
2	A B C	6	M N O
3	D E F	7	P R S
4	G H I	8	T U V
5	J K L	9	W X Y

Many people find it difficult to memorize phone numbers, so they use the correspondence between digits and letters to develop seven-letter words that correspond to their phone numbers. For example, a person whose telephone number is 686-2377 might use the correspondence indicated in the above table to develop the seven-letter word "NUMBERS".

Businesses frequently attempt to get telephone numbers that are easy for their clients to remember. If a business can advertise a simple word for its customers to dial, then, no doubt, the business will receive a few more calls.

Each seven-letter word corresponds to exactly one seven-digit telephone number. The restaurant wishing to increase its take-home business could surely do so with the number 825-3688 (i.e., "TAKEOUT").

Each seven-digit phone number corresponds to many separate seven-letter words. Unfortunately, most of these represent unrecognizable juxtapositions of letters. It's possible, however, that the owner of a barbershop would be pleased to know that the shop's telephone number, 424-7288, corresponds to "HAIRCUT". The owner of a liquor store would, no doubt, be delighted to find that the store's telephone number, 233-7226,

corresponds to "BEERCAN". A veterinarian with the phone number 738-2273 would be pleased to know that the number corresponds to the letters "PETCARE".

Write a C program that, given a seven-digit number, writes to a file every possible seven-letter word corresponding to that number. There are 2187 (3 to the seventh power) such words. Avoid phone numbers with the digits 0 and 1.